

Roll No.

Y – 3958

**MCA (First Semester) EXAMINATION,
Dec.-2021**

Paper –MCA-102

**MATHEMATICAL FOUNDATIONS
COMPUTER SCIENCE**

Time : Three Hours

Maximum Marks : 60

Minimum Pass Marks : 21

Note—Attempt *all* questions. Question no. 1 is compulsory.

1. Attempt any *five* parts— $5 \times 2 = 10$

(i) Define Symmetric relation with example.

(ii) What do you understand by a surjective mapping?

(iii) For every element of $\bar{a}$ of Boolean algebra $[B, +, \cdot, ']$, prove that $a + a = a$
 $\forall a \in B$

ad febe

P.T.O.

- (iv) What do you understand by a statement in logic ? Explain by taking suitable example.
- ~~(v)~~ Define modular arithmetic grammar.
- (vi) Define left and right co-sets of a group.
- (vii) Define irreducible polynomial.
- ~~(viii)~~ Define a field.
- ~~(ix)~~ Define a graph and its applications.
- ~~(x)~~ Define tree and spanning tree.

Unit-I

2. (a) Let R be a binary relation on the set of all positive integer such that : 5
 $R = \{(a, b) : a - b \text{ is divisible by } m, \text{ where } n \in I\},$
 Prove that R is an equivalence relation.
- (b) For any three sets, A , B and C , prove that :

$$A \times (B \cap C) = (A \times B) \cap (A \times C) \quad 5$$

Or

- (a) Apply principle of mathematical induction method to prove that $n^2 - n$ is divisible by 3, where n is a positive integers. 5

(3)

Y - 3958

- (b) Let $D(6)$ is set of elements which are divisible of 6. Then determine $[D(6), 1]$ is a chain. 5

Unit-II

3. Define Boolean algebra. In a Boolean algebra $[B, +, \cdot, ']$ prove that

$$a + b = l. u. b. \{a, b\}$$

$$a \cdot b = g. l. b. \{a, b\}. \quad 10$$

Or

Define logical equivalence of two statements.

Also convert the function

$$f(x, y, z) = (x + y + z) (xy' + x'z)'$$

into conjunctive and disjunctive normal form.

10

Unit-III

4. (a) Prove that intersection of two subgroups of a group G is a subgroup of G . 5

- (b) Prove that H is a normal subgroup of G if and only if $xHx^{-1} = H \forall x \in G$. 5

Or

- (a) State and prove Lagrange theorem for a finite group G . 5

(4)

Y-3958

- (b) If two subgroups H_1 and H_2 of a group G are normal, then prove that $H_1 \cap H_2$ is also a normal subgroup. 5

Unit-IV

5. (a) Find the rational zeros and the decomposition of the polynomial $f(x) = 6x^4 - 7x^3 + 6x^2 - 1$ over the field of rational numbers. 5
- (b) Prove that every field in an integral domain. 5

Or

Show that the set of number of the form $a + b\sqrt{2}$ with a and b are rational numbers is a field. 10

Unit-V

6. Prove that a tree of a connected graph has no circuit. 10

Or

- (a) Explain planar graph and its properties. 5
- (b) Explain Euler graph and its properties. 5

Y-3958

$a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z$

$2x^2 + x + 2$
 $\frac{2x^2 + x + 2}{4x + 4}$
 10

56
 7
 49
 100

$-1x - 1$
 $4x + 4$
 $16x^2 + 4x + 2$
 52